

REPORT FOR LAB1

NAME - SAUMITH DEVARASETTY

Performance Report: Image Mosaic Generator

Method

The mosaic generator reconstructs an input image by dividing it into grid cells and replacing each cell with the closest matching tile from the **CIFAR-100 dataset** (32×32 images). The process involves:

1. **Preprocessing:** Resize and crop the input image to fit the grid.
2. **Grid segmentation:** Compute the mean RGB color of each cell (vectorized NumPy vs. loop).
3. **Tile preparation:** Resize tiles and compute mean LAB color values.
4. **Tile mapping:** Match each cell to the nearest tile based on LAB color distance.
5. **Mosaic output:** Assemble tiles into the final mosaic and evaluate similarity.

Metrics

- **MSE (Mean Squared Error):** Pixel-wise error, lower is better. Sensitive to exact pixel mismatches.
- **SSIM (Structural Similarity Index):** Perceptual similarity (0–1), higher is better. Captures structure, luminance, and contrast.
- **Runtime:** Grid time, mapping time, and total execution time measured in milliseconds.

Results (Similarity + Runtime)

Grid Size	Metric	Value	Grid Time (ms)	Mapping Time (ms)	Total Time (ms)
16×16	SSIM	0.0984	21.8	6159.3	6181.1
	MSE	2429.98	22.2	6091.0	6113.2
32×32	SSIM	0.1370	24.1	1510.4	1534.5
	MSE	3040.36	23.4	1492.0	1515.4
64×64	SSIM	0.2222	22.5	381.0	403.5
	MSE	3322.12	21.7	379.1	400.8

Performance Sweep (Grid Computation Only)

Grid(px)	Vectorized (ms, SSIM)	Loop (ms, SSIM)	Vectorized (ms, MSE)	Loop (ms, MSE)
16	24.5	59.1	22.7	53.5
24	25.8	40.4	20.5	36.2
32	28.7	30.6	19.7	26.8
40	24.0	24.4	19.2	24.2
48	22.8	21.9	19.2	22.5
64	21.8	24.1	19.0	20.4

⚠ *Note: Runtime values may vary slightly between runs due to system load and caching effects, (vectorised faster than loops, convergence at larger grids) remains consistent.*

Analysis

Performance:

- **Mapping is the bottleneck:** At 16×16, mapping alone takes ~6 seconds, while grid computation is only ~20 ms.
- **Vectorized vs. Loop:** Vectorized NumPy is **2–3× faster** at small grids but converges to loop speed at larger grids.
- **Grid size trade-off:**
 - 16×16 → highest detail but very slow.
 - 32×32 → balanced (~1.5s runtime, moderate SSIM).
 - 64×64 → fastest (<0.5s runtime) but produces blockier mosaics.

Metrics:

- **MSE stays high** (2400–3300) due to pixel-level mismatch, making it less useful for mosaics.
- **SSIM improves with larger grids** (0.09 → 0.22), showing better global structure preservation.
- **SSIM is a better metric** for mosaics, as it aligns more closely with human visual perception.

Conclusion

The mosaic generator demonstrates a clear trade-off between **detail and runtime**.

- **Best structural similarity (SSIM = 0.222)** was achieved at 64×64, but at the cost of reduced detail.
- **Best balance** is around **32×32**, where runtime is ~1.5 seconds and similarity remains acceptable.
- **Vectorization is essential** for interactive use, reducing grid computation time by up to 2–3×.
- For quality evaluation, **SSIM is more reliable than MSE**, since it reflects perceptual similarity rather than raw pixel error.